IAA - Deliverable 3
Sueca Game Player

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***Abstract*—This deliverable details the ongoing development of a machine learning model designed to autonomously play the traditional Portuguese card game "Sueca". We outline the data processing pipeline implemented to transform raw match records into turn-by-turn vectors, present our stratified validation strategy, and evaluate baseline and ensemble models including Logistic Regression, Gradient Boosting, and XGBoost.**

***Index Terms*—Machine learning, Sueca, classification, state-of-the-art, data pipeline, gradient boosting, XGBoost.**

I. Introduction

As a quick reintroduction, our project aims to develop a machine learning (ML) model capable of autonomously playing the card game "Sueca". The basic idea is that it is trained in hopes to become good at the decision-making that occurs at a player's turn.

II. State-of-the-Art

Due to the nature of Sueca as a regional card game, there is limited research compared to more widely studied games such as Poker and Bridge. However, the few existing works provide useful insights into possible approaches.

Lopes-Carreiro [1] studies different strategies for Sueca through simulation, comparing heuristic approaches and analyzing their effectiveness.

Correia [2] proposes a Sueca-playing agent integrated into a social robot, using Monte Carlo methods to handle hidden information and incorporating human-robot interaction. These Monte Carlo methods were also a prominent part of the previous paper by Lopes-Carreiro

In a later work, Correia et al. [3] extend this approach by presenting a social robot capable of playing Sueca in real-time scenarios, further exploring the balance between gameplay performance and social interaction.

In a more recent perspective, Torgeir Helgevold [4] discusses the application of machine learning techniques to card games, highlighting how models can learn strategies from gameplay data rather than relying solely on predefined rules or exhaustive search.

From these works, it is evident that Monte Carlo-based methods and heuristic strategies are commonly applied to address uncertainty in Sueca. However, these approaches fall under search-based AI techniques, where decisions are evaluated through simulation rather than learned from data.

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Because with this project we tried to change our vision towards a machine learning paradigm within Artificial Intelligence, we focus on learning decision policies directly from gameplay experience instead of relying directly on search procedures.

III. Data Processing Pipeline

After preparing a method for generating data, which involves using a program that plays and stores data from Sueca games, and due to some choices that were made to increase the performance of the program, we obtained a "working" dataset. However, because of the need to create data fast, said dataset is formatted in a different way than we had expected, and so we also needed to consider a way to process it before feeding it to models.

- **Data aggregation:** Instead of the expected "turn per row" format we wanted, we ended up with an entire round (4 turns) in a single row. While better for generation, this clearly complicates the decision making process, since the first player playing would essentially have to know which cards were going to be played in the future to make a decision. Because of this, we make sure to split each row into 4 individual rows corresponding to the turns in a round.
- **Numeric values:** A lot of the values in the dataset are not numeric (for example, suits appear as their symbol, and player positions appear as cardinals), so a simple conversion of these into numbers was required.
- **Bit encoding:** To facilitate readability, both for us and the model, instead of representing components like cards in the hand and cards played in the trick directly, we use 40-bit vectors, where each bit corresponds to a unique card. For example, in the column `legal_moves_play`, the first play of the game is going to have exactly 30 zeros and 10 ones, which correspond to the 10 cards that the player holds (and is legally allowed to play).

IV. Evaluation Strategy

A. Splitting the Data

We started with a basic Train/Test Split approach, by using 80% of the data for training, and the remaining 20% for validation. However, we figured that we could expand on this a little and secure better results in the future by switching to a Stratified k-fold approach, with $k = 5$. Instead of a single set of training and test data, we split the data into 5 parts (each subset containing 20% of the whole), and train/test each model 5 times, using different combinations of 4 training sets and 1 testing set.

B. Choosing Metrics

Due to the nature of this project as a classification problem, we also started simple, by using metrics such as:

- **Accuracy:** Very simple metric to see the percentage of correct predictions. Not good enough by itself, so other metrics are considered.
- **Log Loss:** Since we are choosing the *one* best card to play, we want to make sure that the model is confident in the card it chose. For that, this metric is helpful.
- **Top-k Accuracy:** Lastly, even though we are trying to maximize the accuracy and confidence of the model, it may also be nice to know if the top-k cards chosen contain the actual best play (starting with $k = 3$). The decision to use Top-k accuracy was done later in the model evaluating phase.

C. Models (Results)

The first values obtained for each of the individual models for comparison were evaluated using 5-Fold Cross-Validation (CV). The consolidated outcomes are presented in Table **??**.

TABLE I
MODEL EVALUATION PERFORMANCE VIA 5-FOLD CROSS-VALIDATION (TRANSPOSED)

Metric	Logistic Reg.	Gradient Boost.	XGBoost
Train Acc.	0.6971	0.9211	0.9323
Val. Acc.	0.6785	0.7978	0.7968
Val. Acc. Std.	$\pm$ 0.0021	$\pm$ 0.0008	$\pm$ 0.0015
Test Acc.	0.6785	0.5769	0.7984

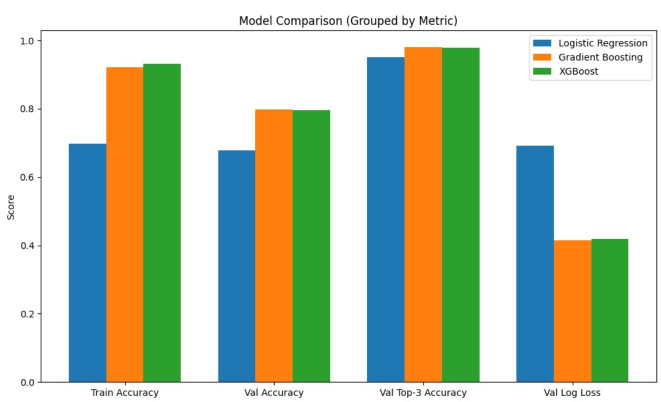

Fig. 1. Graph of the first results obtained

From this, we managed to reach some conclusions:

- Our Logistic Regression model seems to be performing well, but not impressively, as was expected from our choice of a baseline model. Overall it seems to be underfit, and some hyperparameter tuning is going to be done.
- Both Gradient Boosting and XGBoost seem to perform very well on the training data, while dropping in performance when validating with the testing data. This is a clear example of overfitting.
- There seems to be an anomaly with Gradient Boosting, where even though it gets fairly good results, it exhibits a significant drop when evaluated on the unseen holdout dataset.

D. Grid Search Method

To improve our models' performances, we utilized the GridSearchCV class to test each model individually using the same defined metrics. In this part, our choice of parameters to evaluate in each model may not have been as deeply thought out, given time restrictions and the fact that the grid had multiple runs of the models.

- **Logistic Regression:**
Evaluated inverse regularization strengths $C \in \{0.1, 1.0, 10.0\}$ alongside a comprehensive suite of optimization engines including `lbfgs`, `saga`, `liblinear`, `newton-cg`, `newton-cholesky`, and `sag`. This was designed to isolate the optimal weight penalty while auditing convergence speeds across linear solvers.
- **Gradient Boosting:**
Investigated combinations of learning rates ($\eta \in \{0.03, 0.1\}$), structural depth limitations (`max_depth` $\in \{\text{None}, 8, 12\}$), capacity metrics (`max_leaf_nodes` $\in \{31, 63\}$), and terminal node constraints (`min_samples_leaf` $\in \{20, 50\}$). This multidimensional grid aimed to curb the model's tendency to build hyper-isolated, high-variance decision regions.
- **XGBoost:**
Evaluated tree quantities (`n_estimators` $\in \{200, 300\}$) coupled with shallow tree boundaries (`max_depth` $\in \{4, 6\}$) at a fixed learning rate of $\eta = 0.05$. To introduce stochastic regularization, row subsampling and column feature fractions per tree (`colsample_bytree`) were structurally pinned at 0.80 (80%).

After this, we reran the initial model comparison script, applying the new parameters, we ended up with the following results:

TABLE II
POST-OPTIMIZATION PERFORMANCE VIA POST-GRID HYPERPARAMETERS (TRANSPOSED)

Metric	Logistic Reg.	Gradient Boost.	XGBoost
Train Acc.	0.6971	0.8597	0.8841
Val. Acc.	0.6784	0.7978	0.7971
Val. Log Loss	0.6918	0.4381	0.4215
Test Acc.	0.6788	0.7995	0.7973
Test Top-3 Acc.	0.9489	0.9810	0.9797

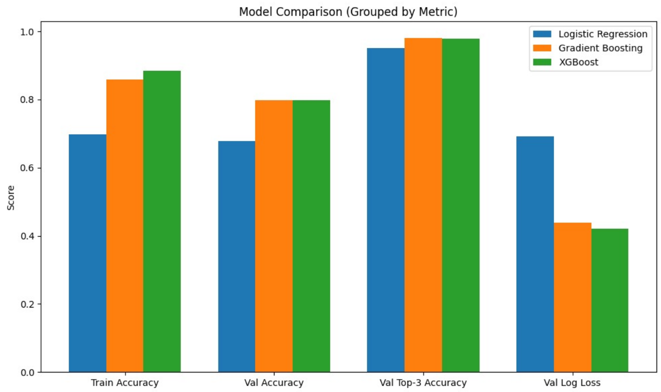

Fig. 2. Graph of the results obtained after the GridSearch

From these new results we have observed and concluded that Logistic Regression is a model that should really only serve as a baseline, given that its underfitting problem has not been mitigated, and its results have barely changed in general. Meanwhile, Gradient Boosting has had a similar improvement to XGBoost, where the Train Accuracy of both has been significantly reduced while the Validation Accuracy has stabilized. While this does not directly mean better results, it does show that the model now learned more generalized strategies, and does not overfit to its training data as much. This is specially relevant for XGBoost, since it is our target for

V. ON-GOING ETHICAL CONSIDERATIONS

Although the development of a machine learning model for something as casual as a traditional card game like "Sueca" may appear to have a small ethical impact at first, there are a lot of important points to consider regarding fairness, data bias and the responsible use of AI methodologies in a card game. These concerns are particularly relevant as the project evolves since it has revealed potential to grow into something that could scale out of the current academic scope in the future.

A. Model Similarity

One of the main ethical considerations to point out relates to the similarity between our model and already existing solutions. Since our dataset is generated through simulated matches of different virtual agents that play a random card from all of the legal plays they can make, there is a risk that the trained model simply reproduces the behavior of those underlying agents rather than genuinely learning optimal strategies from the information he was given.

This raises concerns about the model's originality and transparency. If the model is effectively mimicking the different play patterns that may arise during the data generation process, its performance may be misleadingly interpreted as "intelligent" or "learning", when in reality it is simply replicating prior patterns and assumptions. To mitigate this, we decided that, if further continuity were to be given to training this model, the dataset also include data from matches played by different existing models with heuristic-based decision making, potentially including human game-play data if the scope were ever to grow even further,

Additionally, care must be taken to clearly document the origins of the data and the nature of the agents used to generate it. This ensures that any conclusions drawn about the model's performance are properly contextualized and do not overstate the model's capabilities.

B. Bias and Representativeness

Another relevant concern is the presence of bias in the dataset. Since the data is synthetically generated, it reflects only the strategies and limitations of the programmed agents. This can lead to a narrow representation of game-play styles even when applying the measures mentioned before when talking about model similarity, which could potentially exclude valid but less frequent strategies.

The existence of such bias could result in a model that performs really well in familiar or recurring conditions present in the dataset but fails to properly generalize the situation when exposed to a different or more diverse set of game-play scenarios and/or play-styles. From an ethical standpoint, this limits the robustness and fairness of the model, especially if it were to be used in a competitive or benchmarking environment.

In order to address this, future work should focus on increasing the diversity of game-play data, as mentioned before, either by introducing more varied AI agents or by incorporating human-generated data. This would improve the representativeness of the dataset and reduce the risk of overfitting to a specific style of play.

C. Responsible Use and Scope Limitations

Finally, it is important to consider the intended use of the developed model. While this project is currently purely academic in nature, models capable of playing games autonomously can be extended to competitive environments, potentially raising concerns about the fairness in the usage of the model against human players without proper disclosure.

Although "Sueca" is not typically associated with high-stakes environments, the broader implication is that AI systems should be used transparently and responsibly. Any deployment of the model outside this academic context should clearly indicate that it is an AI agent, ensuring that users are aware of its nature.

Furthermore, the project avoids collecting or using personal data, minimizing privacy concerns. However, maintaining this principle in future extensions such as incorporating human game-play data will be essential to ensure compliance with ethical and data protection standards.

D. Conclusion

Overall, while the ethical risks in this project are relatively low, adopting responsible practices at this stage establishes a strong foundation for the possibility of any future development taking place and aligns with best practices in machine learning research.

VI. Critical Reflection

Looking back at the development of this project, several important insights can be drawn regarding both technical and conceptual decisions made throughout the process.

One of the main challenges encountered was related to the data generation pipeline. In order to quickly obtain a sufficiently large dataset, we opted for a simulation-based approach using virtual agents. Although this allowed us to scale data generation efficiently, it introduced structural issues in the data set, such as the aggregation of multiple turns into a single row. This required additional pre-processing steps and ultimately highlighted the importance of designing data formats with the final learning task in mind from the beginning.

Another key realization concerns the trade-off between model complexity and generalization. While more advanced models such as Gradient Boosting and XGBoost achieved very high training accuracy, they also exhibited clear signs of overfitting when evaluated on unseen data. This emphasizes that better performance on training data does not necessarily translate into better real-world decision-making and that careful validation strategies are essential in machine learning workflows.

Additionally, the choice of evaluation metrics proved to be more significant than initially expected. Relying solely on accuracy would have provided an incomplete picture of model performance. The inclusion of metrics such as Log Loss and Top-k Accuracy allowed for a more nuanced understanding, particularly in a setting where multiple decisions may be considered reasonable rather than strictly correct or incorrect.

Finally, given more time, several improvements could be explored. These include generating higher-quality and more diverse datasets, incorporating human game data, applying more robust regularization and hyperparameter tuning techniques, and exploring alternative model architectures. Such enhancements would likely improve both the performance and reliability of the system.

Overall, this project provided valuable hands-on experience with the full machine learning pipeline, from data generation and pre-processing to model evaluation and critical analysis. It also reinforced the importance of iterative development and continuous validation when designing intelligent systems.

VII. AI Tools

When it comes to AI usage, there were three distinct places that warrant mentioning:

- **Data generation/processing:**
  For generating data, as it was outside of our intended scope, and seeing as we wanted more time to ponder the data, we used GitHub Copilot to help create code that simulated games and stored data in a format we believed to be useful (which still required processing as we realized some fundamental flaws). During said processing of this odd data, we used Perplexity to help create the script cleans the data and restructures it.
- **Baseline model comparison code:**
  To define the simplest way for a comparison of the 3 models of choice, we asked Gemini to setup a very simple display of the data, given the cross-validation and metrics of our choosing.
- **IEEE document formatting:**
  We started writing this document in a standard manner, but when we realized that the required format was IEEE, we used Gemini to create the document template using the proper format.

VIII. References

References

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